

CHEMICAL ANALYSIS – INSTRUMENTAL METHODS

KEY VOCABULARY

Instrumental method: uses machines to analyse substances.

Flame emission spectroscopy: identifies metal ions from a light spectrum.

Accurate: results are close to the true value.

Sensitive: can detect very small amounts.

COMMON MISCONCEPTIONS

~~— Instruments are slower than test tubes.~~

✓ They are usually faster and more accurate.

~~— They need large samples.~~

✓ They are sensitive — tiny samples work.

~~— Flame tests are more reliable for mixtures.~~

✓ Instruments handle mixtures far better.

WHY USE INSTRUMENTS

Faster than manual chemical tests.

More accurate and more sensitive.

Can analyse very small samples.

Can tell apart ions a flame test confuses.

Flame emission spectroscopy → metal ions.

1

DEFINE

State what is meant by an **instrumental method**. [1 mark]

2

ADVANTAGES

Give three advantages of instrumental methods over chemical tests. [3 marks]

3

SPECTROSCOPY

Name the instrumental method used to identify **metal ions**. [1 mark]

4

WHEN USEFUL

Suggest why instruments are useful for analysing a tiny crime-scene sample. [2 marks]

5

FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Instrumental methods are fast, _____ and very _____.

accurate

sensitive

slow

large

6

INTERPRET

Explain how flame emission spectroscopy identifies an ion. [2 marks]